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Senior Project I

# 1. Plan Introduction

This software development plan outlines the approach, resources, organization, and schedule for **Pathfinder**, an all-in-one platform designed to guide users through their educational and professional journeys. The project aims to deliver a comprehensive toolset for students, graduates, and professionals, offering personalized support in career guidance, resume building, job matching, and networking, with the assistance of an AI-powered chatbot.

## 1.1.1 Project Deliverables

- **Platform Design Document (PDD):** A detailed document outlining Pathfinder's core features, workflows, and user interface.
- **Alpha Version:** An initial version of Pathfinder, featuring the basic career questionnaire, resume analysis, and cover letter generator tools.
- **Beta Version:** A feature-complete version of the platform, including job matching, learning resources, networking tools, and AI chatbot integration.
- **Release Version:** Final version of Pathfinder, incorporating user feedback, bug fixes, and optimization for the platform's launch.

## 1.2 Project Resources

### 1.2.1 Hardware Resources

- **Development Computers:** High-performance PCs capable of running development tools and handling cloud services for AI integration.
- **Testing Devices:** Various devices such as Windows/Mac laptops, Android/iOS smartphones for cross-platform testing and responsiveness.

### 1.2.2 Software Resources

- **Web Development Frameworks:** React.js for front-end development, Node.js for back-end.
- **Databases:** FireBase for storing user data and system records.
- **AI Integration Tools:** OpenAI API for chatbot functionality.
- **Version Control System:** Git for source code management, with GitHub as the repository for collaboration.
- **Design Tools:** Figma or Adobe XD for UI/UX design.
- **Cloud Hosting:** AWS or Google Cloud for server hosting, database management, and AI services integration.

# 1.3 Project Organization

### Roles:

- **Front-End Developer:** Responsible for developing the user interface, ensuring responsiveness, and integrating with the backend.
- **Back-End Developer:** Focuses on server-side logic, database management, API development, and integrating AI chatbot functionalities.
- **UI/UX Designer:** Designs the interface and ensures a user-friendly experience across all features.
- **Project Manager:** Ensures timely completion of tasks, coordination among team members, and maintaining the overall project timeline.

### Collaboration:

- Weekly team meetings will be held to review progress, discuss blockers, and set short-term goals. The project manager will facilitate these meetings and ensure alignment on priorities.

# 1.4 Project Schedule

### 1.4.1 PERT / GANTT Chart

A GANTT chart will be created using project management software such as Microsoft Project or Jira, to visualize the project timeline, highlighting key milestones, dependencies, and deadlines.

### 1.4.2 Task / Resource Table (Note: Some Overlap)

| Task | Resource | Duration | Dependencies |
|---|---|---|---|
| Platform Design Document (PDD) | Entire team | 2 Weeks | None |
| Front-End Development | Front-End Developer | 2 Weeks | PDD |
| Back-End Development | Back-End Developer | 2 Weeks | PDD |
| Career Questionnaire | Entire team | 1 Weeks | PDD |
| Resume Analysis Tool | Back-End Developer | 2 Weeks | Back-End Development |

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|---|---|---|---|
| Cover Letter Generator | Back-End Developer | 1 Weeks | Resume Analysis Tool |
| Job Matching Feature | Back-End Developer | 2 Weeks | Front-End, Back-End |
| Learning Resources | Entire team | 1 Weeks | Front-End, Back-End |
| Networking Tools | Entire team | 1 Weeks | Front-End, Back-End |
| Alpha Version | Entire team | 1 Weeks | Resume, Cover Letter Tools |
| Beta Version | Entire team | 1 Weeks | Alpha Version |
| Testing & Feedback | Entire team | 1 Weeks | Beta Version |
| Final Adjustments | Entire team | 1 Weeks | Testing & Feedback |
| Launch Preparation | Entire team | 1 Weeks | Final Adjustments |
| Post-Launch Support | Entire team | Ongoing | Launch |